

Report: Applications of Data Science in E-commerce

1. Introduction

E-commerce platforms generate massive volumes of data from customer interactions such as browsing behavior, purchases, reviews, payments, and logistics. Data Science helps convert this raw data into meaningful insights for better decision-making, improved customer experience, and optimized operations.

2. How Data Science is Used in E-commerce

Key applications include personalized recommendations, customer segmentation, demand forecasting, dynamic pricing, fraud detection, and customer churn prediction. These applications improve satisfaction, conversion rates, and average order value while reducing costs.

3. Customer Segmentation in E-commerce

Customer segmentation groups customers based on demographic, behavioral, and value-based data. It enables targeted marketing, personalization, and customer retention.

4. K-Means Clustering

K-Means is an unsupervised machine learning algorithm used to group customers into clusters based on similarity. It is widely used for customer segmentation and targeted marketing strategies.

5. Tools and Methods

Tools include Excel, SQL, Python, Power BI, Tableau, and Google Analytics. Methods include descriptive, diagnostic, predictive, and prescriptive analytics.

6. Case Studies

Amazon uses recommendation systems contributing nearly 35% of its sales. Flipkart uses predictive analytics and sentiment analysis to improve logistics and customer satisfaction.

7. Business Questions

Key questions include product recommendations, churn prediction, pricing optimization, and campaign ROI measurement.

8. Conclusion

Data Science is a core driver of e-commerce success, enabling personalization, efficiency, and data-driven growth.